Investigating Max Pressure Jump in Mixed Phase*

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We investigate the first-order phase transition between hadronic and quark matter in the core of neutron stars by using a quadratic interpolation method. Our approach modifies the Maxwell construction to include an additional pressure term that accounts for structural effects in the mixed phase, often referred to as the "pasta phase."[1] By implementing a piecewise polytropic equation of state (EOS) for the hadronic and quark phases, we create a smooth transition across the mixed phase[2]. This model enables us to examine the maximum achievable pressure jump while maintaining continuity and stability between the phases. We explore how varying parameters, specifically the polytropic index Γ and critical chemical potential $\mu_{\rm crit}$, affect the phase transition properties. This approach allows us to see how the mixed phase changes with respect to different EOS.

I. INTRODUCTION

The transition from the hadronic to quark phase remains incomplete in the low temperature and high density region. The transmission can be either discontinuous or smooth. The study of high-density matter is essential for understanding extreme states that cannot be recreated on Earth yet may occur naturally within the cores of neutron stars. Neutron stars, with densities reaching several times that of nuclear matter, provide a unique environment for such research. The extreme pressures and densities in these compact stars widely expect the emergence of a quark phase, where quarks no longer confine within protons and neutrons, but instead form a distinct state of matter known as quark matter. Theoretically, at sufficiently high densities, up to a few times nuclear saturation density, this transition to a quark phase occurs [3].

The phase transitions within neutron stars particularly the transition from a hadronic phase to a quark matter phase are of considerable interest. Under these extreme conditions, we expect a first order phase transition, known as the hadron-quark transition. Within this transition, the "pasta phase" may arise as a mixed phase where hadrons and quarks coexist [4]. The pasta phase smoothens the equation of state (EOS) by bridging the two phases, mitigating abrupt changes in density. However, this approach presents challenges in maintaining smoothness within the EOS, an issue we address in this work. We use a quadratic interpolation method with an extra pressure term to account for the effects of the structure on the critical pressure in the Maxwell construction [2] to get a rough idea of the pasta phase without explicitly computing the geometrical structures.

We are well aware of the properties of the equation of state around the nuclear saturation density, n_0. By setting the values of μ and P at this saturation density, we want an equation that gives us the minimum possible value for the additional pressure term for a given P and μ from which we can have the maximum smoothness an EOS can have when we are given two EOS that have a first-order phase transition. If we can find an expression for the additional pressure term, we can apply it to any given equation of state (EOS).

II. MOTIVATION

Understanding the structure and stability of hybrid compact stars through the investigation of their mixed phases is crucial. Therefore, the development of a mathematical treatment to study these phase transitions is necessary. However, the lack of well-studied microphysics of these transitions limits our understanding. Addressing these gaps is crucial for accurately describing matter in such extreme environments and connecting theory with astrophysical observations. In this study, we aim to investigate the properties of neutron stars based on their degree of smoothness, beginning with a first-order phase transition.

III. FORMALISM

First, we use piecewise polytrope to build EOS for the hadronic and quark phases. Next, we search for a quadratic interpolation curve that can replace the mixed phase, exhibits the highest smoothness, and traverses various critical pressure values, specifically the intersection pressure of the hadronic and quark polytropes.

A. PIECEWISE POLYTROPE EOS WITH A FIRST- ORDER PHASE TRANSITION

To investigate the possibility of describing compact star phenomena using a piecewise polytropic EOS at supersaturation densities, we define one of the polytropes as a constant pressure region with $P = P_{\rm crit}$, where

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a jump in energy density ϵ due to a first-order phase transition fulfills the Seidov constraint [5].

$$\frac{\Delta\epsilon}{\epsilon_{\rm crit}} \geq \frac{1}{2} + \frac{3}{2}\frac{P_{\rm crit}}{\epsilon_{\rm crit}}, \tag{1}$$

The supersaturation density region is divided into three polytropic regions where n_1 just above the nuclear saturation density ($n_o = 0.15\,\mathrm{fm}^{-3}$) and polytrope equation fulfill

$$P(n) = k(n)^{\Gamma} \tag{2}$$

$$\begin{aligned} i = 1 &: n_1 \geq n \geq n_{12} \\ i = 2 &: n_{12} \geq n \geq n_{23} \\ i = 3 &: n \geq n_{23} \end{aligned} \tag{3}$$

For the above regions at zero temprature we use the known thermodynamical relations for maxwell construction of a first order phase transition [6],

$$P(n) = n^2 \frac{d\left(\frac{\varepsilon(n)}{n}\right)}{dn}, \tag{4}$$

$$\frac{\varepsilon(n)}{n} = \int dn \frac{P(n)}{n^2} = \int dn\, \kappa n^{\Gamma-2} = \frac{\kappa n^{\Gamma-1}}{\Gamma-1} + C, \tag{5}$$

Here the integration constant C will be m_0, from the condition $\varepsilon = m_o n$ at $n \to 0$

$$\mu(n) = \frac{P(n)+\varepsilon(n)}{n} = \frac{\kappa\Gamma}{\Gamma-1} n^{\Gamma-1} + m_0, \tag{6}$$

$$n(\mu) = \left[(\mu - m_0)\frac{\Gamma-1}{\kappa\Gamma}\right]^{\frac{1}{\Gamma-1}}, \tag{7}$$

$$P(\mu) = \kappa\left[(\mu - m_0)\frac{\Gamma-1}{\kappa\Gamma}\right]^{\frac{\Gamma}{\Gamma-1}}. \tag{8}$$

B. EQUATION OF STATE

From the above relations we construct EOS with a phase transition. We have divided i = 3 into two branches i = 3a and i = 3b their respective density range will be $n_{23} \geq n \geq n_{34}$ and $n \geq n_{34}$ where $n_1 = 0.165 fm^{-3}, n_{12} = 0.32 fm^{-3}, n_{23} = 0.5344 fm^{-3}, n_{34} = 0.7500 fm^{-3}$

The EOS has following set of properties :

$$P_{\rm crit} = 63.177\,\mathrm{MeV\ fm}^{-3},$$

$$\varepsilon_{\rm crit} = 318.26\,\mathrm{MeV\ fm}^{-3},$$

$$\Delta\varepsilon = 253.89\,\mathrm{MeV\ fm}^{-3}.$$

In the FIG. 1 the intersection point of hadronic and quark phase represents the critical values of pressure and chemical potential, we can find the critical value of energy density from the specified relations. The $\Delta\varepsilon$ represents the jump in the energy density of mixed phase region the same we can see in the FIG. 2.

TABLE I: Values of Γ_3, κ_3, and m_0 for different parameter sets

	Γ	κ [MeV fm$^{3(\Gamma-1)}$]	m_0 [MeV]
i = 1	4.921	17906.60	939.56
i = 2	0	63.177	1201.75
i = 3a	4.00	774.375	1031.815
i = 3b	2.80	548.309	958.553

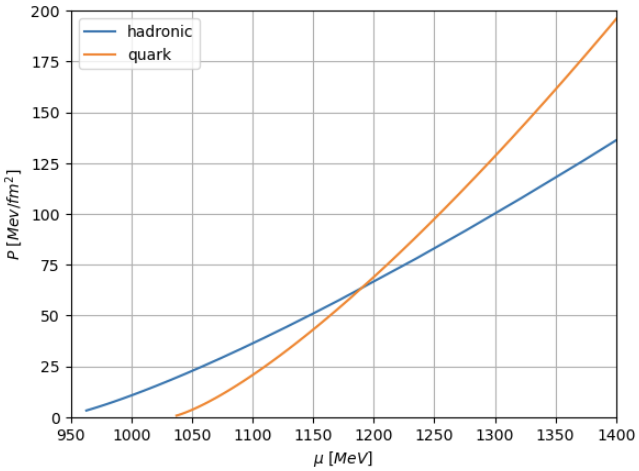


FIG. 1: Pressure vs chemmical potential of the piecewise polytropes.

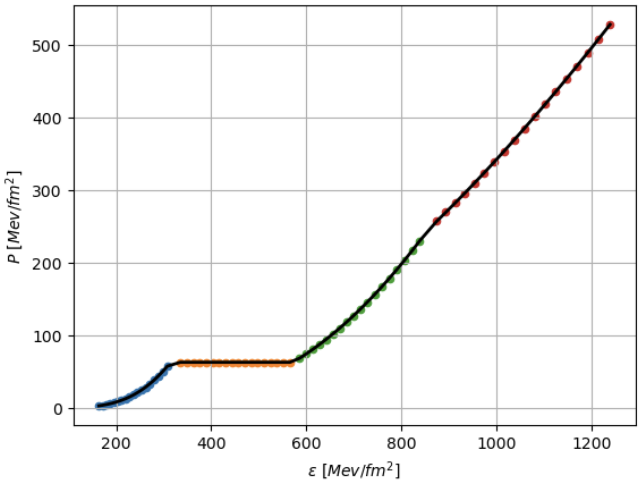


FIG. 2: Pressure vs Energy density of piecewise polytropes showing different phases.

C. QUADRATIC INTERPOLATION FOR THE FIRST- ORDER PHASE TRANSITION

In the FIG. 3 $P_H(\mu), P_Q(\mu)$ represents the Pressure of the hadronic and quark phase dependence in terms of chemical potential respectively. P_C be the critical pressure where both the hadronic and quark EOS have same chemical potential, we can write P_c in terms of

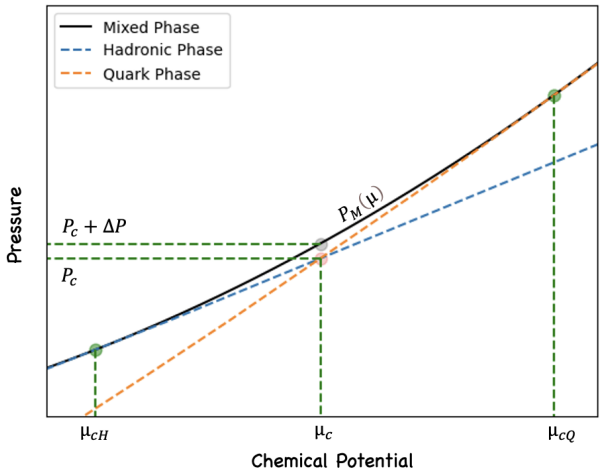


FIG. 3: The schematic representation of the interpolation function $P_M(\mu)$, derived from the mixed phase constructions discussed in this work.

$P_H(\mu), P_Q(\mu)$ as,

$$P_H(\mu_c) = P_Q(\mu_c) = P_C \tag{4}$$

We can modify the maxwell construction on the assumption of both the EOS are changing due to their interactions.Now we will incorporate an additional pressure term ΔP which will be added to the critical pressure P_c having condition that the mixed phase equation should pass through this additional pressure, so we construct an polynomial around this point,

$$P_M(\mu) = \alpha(\mu - \mu_c)^p + \beta(\mu - \mu_c)^q + P_c + \Delta P \tag{5}$$

As this study is mostly focused on quadratic interpolation the p,q will take the values 2 and 1 in the equation,

$$P_M(\mu) = \alpha(\mu - \mu_c)^2 + \beta(\mu - \mu_c) + P_c + \Delta P \tag{6}$$

Here we are constructing a quadratic equation (Mixed Phase) around the point $P + \Delta P$ such that the change in hadronic Phase to mixed phase and mixed phase to quark phase is smooth without any jump in the density The number density $n(\mu)$ is given by the derivative of the pressure with respect to the chemical potential:

$$n(\mu) = \frac{dP(\mu)}{d\mu}.$$

Thus, there are two critical chemical potentials: μ_{cH} for the transition from the hadronic phase to the mixed phase and μ_{cQ} for the transition from the mixed phase to the quark phase. As a result, there are four unknowns, including the coefficients α and β.The phase transition conditions can be written as [2]:

$$\begin{cases} P_M(\mu_{cH}) = P_H(\mu_{cH}), \\ P_M(\mu_{cQ}) = P_Q(\mu_{cQ}), \end{cases} \quad \begin{cases} n_M(\mu_{cH}) = n_H(\mu_{cH}), \\ n_M(\mu_{cQ}) = n_Q(\mu_{cQ}). \end{cases}$$

If we solve for these set of equations, we obtain the equations for α and β,

$$\begin{cases} \alpha = \frac{1}{\mu_{cQ}-\mu_{cH}} \left(\frac{P_Q(\mu_{cQ})-(\Delta P+P_c)}{\mu_{cQ}-\mu_c} - \frac{P_H(\mu_{cH})-(\Delta P+P_c)}{\mu_{cH}-\mu_c} \right) \\ \beta = \frac{P_H(\mu_{cH})-(\Delta P+P_c)}{\mu_{cH}-\mu_c} + \frac{P_Q(\mu_{cQ})-(\Delta P+P_c)}{\mu_{cQ}-\mu_c} - \frac{P_Q(\mu_{cQ})-P_H(\mu_{cH})}{\mu_{cQ}-\mu_{cH}}, \end{cases} \tag{7}$$

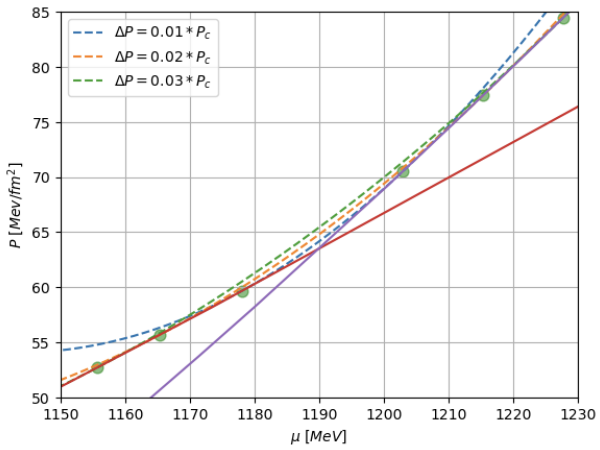


FIG. 4: Incorporation of additional pressure term in the mixed phase.

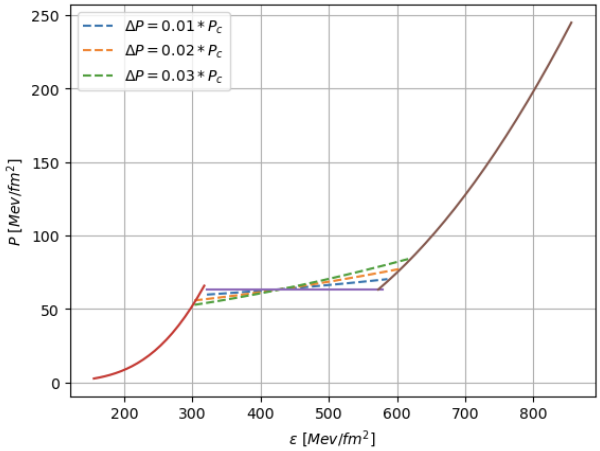


FIG. 5: Change in the Pressure vs Energy density plot due to additional pressure term.

D. MAXIMUM INTERPOLATION ACHIEVABLE

Now if we fix the μ_{cH} value to the initial value of the hadronic phase, we desire an μ_{cQ} such that the mixed phase EOS passes through both of the points and it should also pass through any of the additional pressure points, i.e., $P_c + \Delta P$ ($\Delta P \in [0.1, 0.9]P$) .Now we will develop a model that will determine the mixed phase equation of state (EOS) by imposing the following conditions the output of the model will give us the additional pressure term within the pressure limit along with the μ_{cQ} value:

Equality Constraint

We fix the initial value of the chemical potential in the hadronic phase to be our μ_{cH}, to serve as a boundary condition for the mixed phase EOS thus requiring the mixed phase EOS, to always pass through the initial pressure and chemical potential which will also become our P_{cH} , μ_{cH} . This ensures that the EOS is consistent with the hadronic phase at the starting point.

Inequality Constraint

In addition to the equality constraint, we impose an inequality constraint on the mixed phase EOS. Specifically, the EOS curve $P_M(\mu)$ should remain above or intersect with the EOS of the hadronic and quark phases. This ensures that the mixed phase EOS stays physically consistent with both phases and reflects the transition between them.

$$P_M(\mu) \geq P_H(\mu) \quad \text{and} \quad P_M(\mu) \geq P_Q(\mu),$$

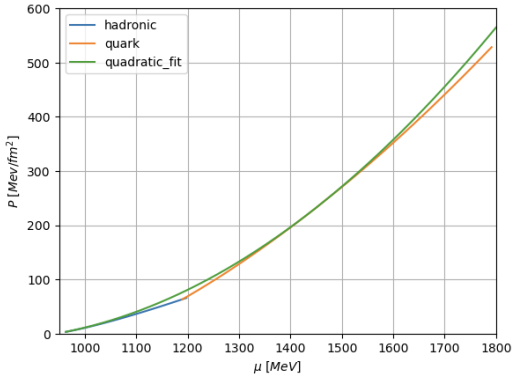


FIG. 6: The schematic representation of interpolating the quadratic fit for the given EOS.

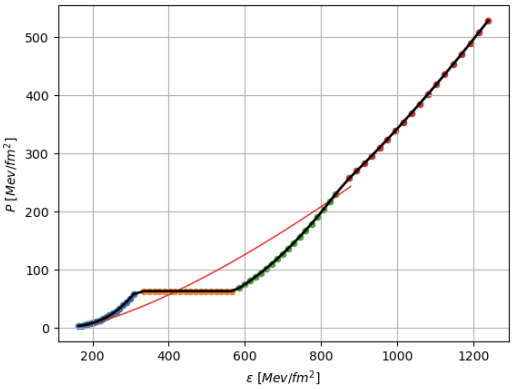


FIG. 7: Incorporation of mixed phase in between of hadronic phase and quark phase in Pressure vs Energy density plot.

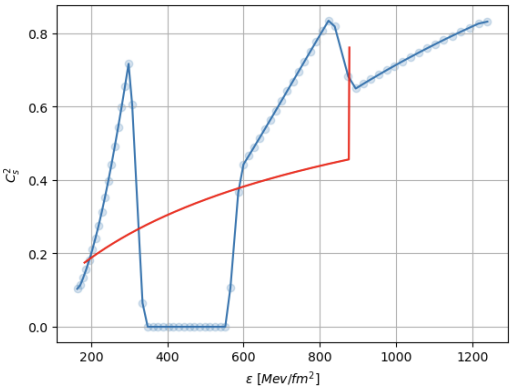


FIG. 8: This graph demonstrates effect of quadratic fit on speed odyssey sound vs energy density plot.

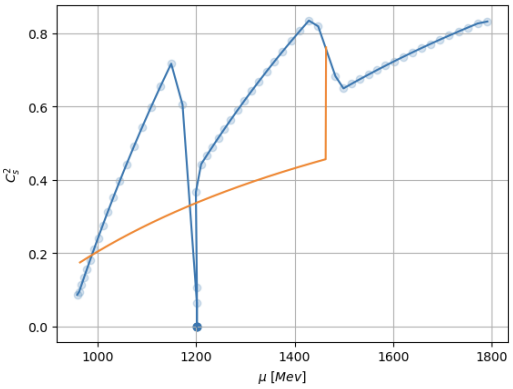


FIG. 9: This graph demonstrates the effect of quadratic fit on speed of sound vs chemical potential plot.

Optimization Technique

To determine the optimal values of the coefficients α and β in the quadratic equation for $P_M(\mu)$, we employ an optimization technique. The objective is to ensure that

the quadratic fit follow the given conditions.

The quadratic function should satisfy the following conditions:

- It must pass through the initial chemical potential and pressure point of the hadronic phase, ensuring continuity with the hadronic phase.
- It must intersect the quark phase EOS, $P_Q(\mu)$, at the quark EOS chemical potential μ_{cQ}.
- The mixed phase EOS stay above both the hadronic and quark curves.

Thus, the algorithm[7] incorporates both equality and inequality constraints to ensure the mixed phase EOS is physically feasible. The equality constraints enforce that the quadratic function passes through the initial hadronic phase point, while the inequality constraints ensure the EOS stays above the hadronic and quark phases.

4. Model Outputs

The optimization model will predict the optimal values of the coefficients α and β for the quadratic equation. Additionally, the model will provide the optimal pressure values $P_c + \Delta P$, ensuring that the mixed phase EOS satisfies the necessary physical conditions. The model will also output the optimal quadratic fit, $P_M(\mu)$, which will respect the constraints and ensure the transition between phases is correctly captured.

IV. RESULTS

Relation of optimal pressure to Different Γ_Q.

By modifying the values of Γ of Quark in the equation of state (EOS), we generate an alternative quadratic relationship, which offers an opportunity to investigate how different gamma quark values impact the characteristics of the phase transition. Specifically, adjusting Γ_Q influences the stiffness and overall shape of the EOS, thereby altering the predicted behavior of mixed phase pressure as a function of μ. This model deepens our understanding of the mixed phase that governs the transition between hadronic and quark phases in neutron star matter.

From the FIG. 10 we can see that the down trend in the P+ΔP values as we keep on increasing Γ_{Quark}.

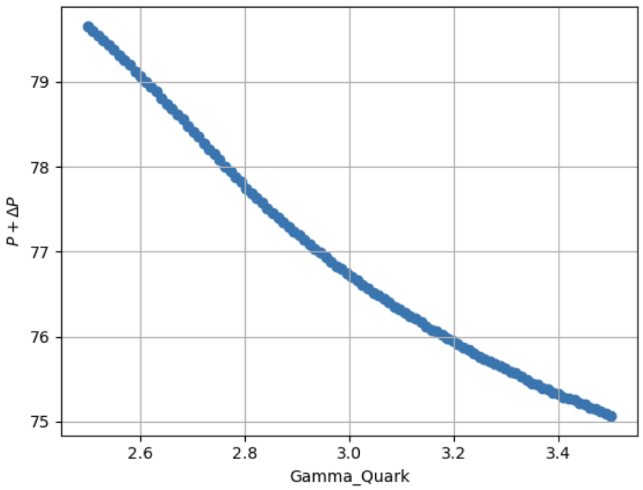


FIG. 10: Relation Between Γ_Q and $P_c + \Delta P$

V. FUTURE WORK

In this work, we address the challenge of achieving smoothness in the pasta phase, which is expected to arise at high densities. We investigate the maximum smoothness achievable when using a polytrope to model the hadronic phase and another to model the quark phase. I plan to extend this study by varying the critical chemical potential, μ_c, which shifts the onset densities and allows exploration of its effects on the smoothness parameter, ΔP and also understanding the physical interpolation with respect to the EOS of Neutron Stars.

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